MODULE 04 · EXECUTION

Options Basics

An option is a right, not a thing you own. That single shift — plus a little vocabulary about time and probability — explains why option prices behave so differently from the stock.

BEGINNER

25 MIN READ

“Discipline before conviction.”

BY THE END, YOU WILL BE ABLE TO

01
Know what a **call and a put** actually represent

02
Read **premium, the Greeks & moneyness** at a beginner level

03
Understand **why cheap contracts aren’t cheaper**

01 A RIGHT, NOT AN OBLIGATION

An option is a contract about a future price

Strip away the jargon and an option is simple: you pay a fee today for the *right* to trade something at a fixed price, up to a deadline.

A single option contract gives its owner the **right — but never the obligation —** to buy or sell 100 shares of a stock at a fixed price (the **strike**), any time until a set date (**expiration**). For that right, you pay a one-time fee called the **premium**.

Strike
The fixed price you can transact at.

Expiration
The deadline the right runs out.

Premium
The fee you pay for the right.

Contract
Covers 100 shares of stock.

■ AN ANALOGY

Think of a refundable deposit on a house at a locked-in price. If the market price soars, your locked price is gold — you profit. If it falls, you simply walk away, losing only the deposit. The deposit is the **premium**; the locked price is the **strike**.

◆ KEY IDEA

Because it's a **right, not an obligation**, the most a buyer can lose is the premium — while the upside can be far larger. That asymmetry is the whole appeal, and the whole trap.

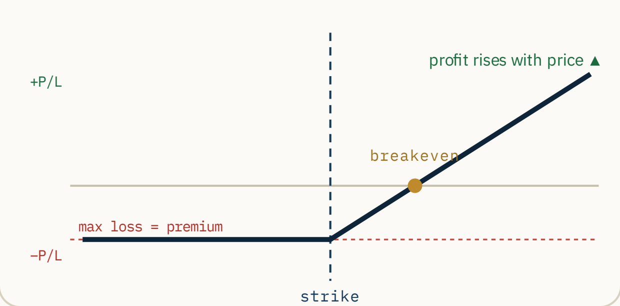
One contract represents **100 shares**, so a premium quoted at `$2.10` actually costs `$210` to buy. Keep that ×100 in mind — it surprises every beginner once.

02 TWO BETS, MIRRORED

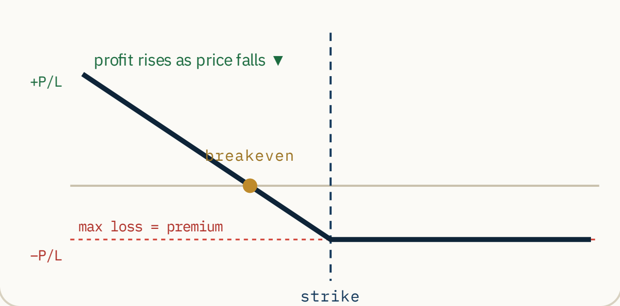
Calls, puts & the payoff picture

There are only two option types. A **call** profits when price goes up; a **put** profits when price goes down. These payoff diagrams show the whole story.

LONG CALL — betting price RISES



LONG PUT — betting price FALLS



Call — the right to **buy** at the strike. You want price *above* the strike. Loss is capped at the premium; profit grows as price climbs.

Put — the right to **sell** at the strike. You want price *below* the strike. Same capped loss; profit grows as price falls.

◆ KEY IDEA — BREAKEVEN

You don't profit the instant you're right on direction — you have to first earn back the premium. **Call breakeven = strike + premium. Put breakeven = strike − premium.** Below/above that, a "correct" call/put can still lose.

(There is also *selling* options, which flips this risk profile entirely — capped gain, large risk. That's an advanced topic for later; here we only buy.)

03 INSIDE THE PREMIUM

Premium = real value + time value

Every premium splits into two parts. Knowing which part you're buying explains most of why options gain or lose value when you least expect it.

Intrinsic value is the part that would already be worth something if the option expired right now — the real, in-the-money portion. **Extrinsic value** (time value) is everything else: what you pay for the *possibility* that price keeps moving your way before expiration.

IN-THE-MONEY CALL — premium $5.20

intrinsic $3.00 | time value $2.20

OUT-OF-THE-MONEY CALL — premium $0.80

time value $0.80

The out-of-the-money contract is **100% time value** — you're paying entirely for hope. That portion melts away as expiration approaches (next page). The in-the-money contract has a solid base of real value that doesn't evaporate the same way.

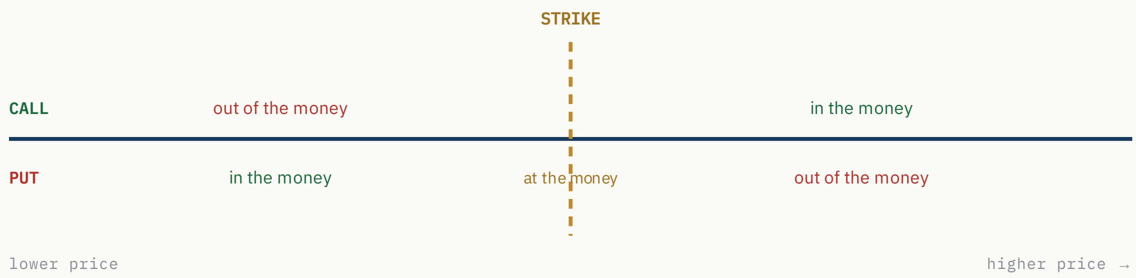
▲ WHY YOUR OPTION CAN FALL WHILE THE STOCK RISES

If you own a call and the stock ticks up *slowly*, the time value can bleed out faster than the small move adds intrinsic value — so the option loses money even though you were "right." Options price **movement and time**, not just direction.

04 WHERE PRICE SITS VS THE STRIKE

In, at & out of the money

“Moneyness” is just the relationship between the current price and the strike — and it flips depending on whether you hold a call or a put.



- **In the money (ITM):** the option already has intrinsic value. A call is ITM when price is *above* the strike; a put when price is *below* it.
- **At the money (ATM):** price sits right at the strike — nearly all time value, and the most sensitive to movement.
- **Out of the money (OTM):** no intrinsic value yet — pure hope. Cheapest, and the most likely to expire worthless.

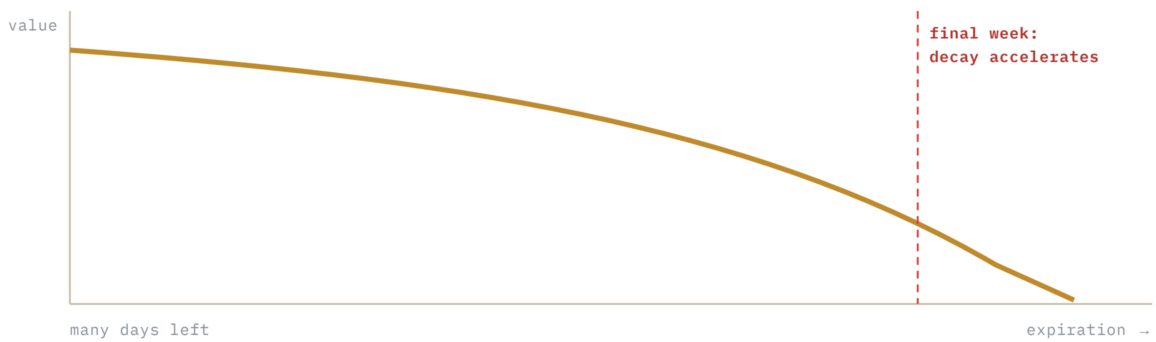
■ WHY IT MATTERS

Moneyness sets the trade-off you’re choosing: ITM costs more but moves closely with the stock; OTM is cheap but needs a big, fast move just to survive. The next pages are about that trade-off.

05 THE MELTING ICE CUBE

Expiration & time decay (theta)

Time value doesn't drain evenly. It bleeds slowly at first, then collapses in the final stretch before expiration — whether or not the stock moves.



Theta is the name for this daily bleed of time value. An option is a little like a melting ice cube: hold it too long and there's nothing left, even if you were eventually right on direction. Weekends melt too — the clock doesn't pause when the market closes.

◆ THE TRADE-OFF

More time = more premium but slower decay. Less time = cheaper but brutal decay. Very short-dated options are mostly a bet on a move happening **now**.

▲ THETA WORKS AGAINST BUYERS

Every day you simply *hold* a bought option, time decay quietly charges you rent. Direction has to outrun that rent to win.

06 THREE DIALS, BEGINNER LEVEL

The Greeks, without the calculus

The “Greeks” measure what moves an option’s price. You only need three to start — think of them as dials, not equations.

δ

DELTA — DIRECTION

Roughly how many dollars the option moves per $1 move in the stock. ~0.50 ATM, higher ITM, lower OTM. Loosely, a rough sense of the odds of finishing ITM.

θ

THETA — TIME

The daily cost of time decay you met on the last page. For a buyer it’s negative — value lost just by the clock ticking.

ν

VEGA / IV — VOLATILITY

Sensitivity to **implied volatility** (IV) — the market’s expectation of movement. Higher IV inflates every premium.

▲ THE IV TRAP — “IV CRUSH”

Before big events (like earnings) IV spikes, making options expensive. After the news, IV collapses — and premiums can **drop hard even if the stock moved your way**. Buying high-IV options into an event is a classic beginner loss.

There are more Greeks (gamma, rho…), but delta, theta, and IV explain most of what a beginner will actually feel. Master the intuition first; the rest is refinement.

07 THE MOST EXPENSIVE BEGINNER HABIT

Why a cheap contract isn’t a cheaper bet

The $0.20 lottery-ticket option looks like a bargain next to the $4.00 one. On the math, it’s usually the worse deal — here’s why.

	Far-OTM — $0.20 the “cheap” one	Near-money — $4.00 the “expensive” one
What you’re buying	100% time value (hope)	Mostly real, intrinsic value
Delta (moves with stock)	Tiny — barely reacts	Strong — tracks the stock
Move needed to win	A big, fast one	A modest one
Time decay	Vicious — bleeds to zero	Cushioned by intrinsic value
Most likely outcome	Expires worthless	Survives small moves

“Cheap” only describes the sticker price, not the **odds**. Buying ten $0.20 contracts feels thrifty, but you’ve bought ten near-coin-flips that decay fast and need a large move just to break even. Price per contract is not the same as cost of being wrong.

◆ KEY IDEA

Don’t shop for the lowest premium. Shop for the contract whose **delta, time, and breakeven** actually fit the move you expect. A more expensive option is often the cheaper way to be right.

08 DECODE THE TICKET

Reading an option, field by field

Put it all together by decoding a single order ticket the way you'd read a sentence.

SPY 580 C 05/29 @ $2.10

SPY — underlying
The stock/ETF the option is written on.

580 — strike
The fixed price you have the right to buy at.

C — call
A call (right to buy). 'P' would be a put.

05/29 — expiration
The right expires after this date.

SO WHAT HAS TO HAPPEN?

This contract costs **$2.10 × 100 = $210**. It's a bet SPY closes **above $580** by May 29. But you don't profit at $580 — you profit above **breakeven $582.10** (strike + premium). Between now and then, **time decay** charges rent daily. Your max loss is the full $210; your upside grows the further past $582.10 SPY runs.

THE READING HABIT

Before any options trade, narrate the ticket out loud: underlying, strike, type, expiry, premium, breakeven. If you can't say what has to happen **and by when**, you're not ready to click.

09 LOCK IT IN

What to carry forward

Six ideas that turn options from a mysterious lottery into a tool you can reason about — or knowingly leave alone.

1. An option is a **right, not an obligation**; max loss is the premium.
2. **Calls** profit up, **puts** profit down — past breakeven, not the strike.
3. Premium = **intrinsic + time value**; OTM is all hope.
4. **Theta** bleeds time value, fastest near expiration.
5. **Delta, theta, IV** explain most of what you'll feel.
6. **Cheap isn't cheaper** — match the contract to the move you expect.

A–Z QUICK GLOSSARY

Call / Put
Right to buy / right to sell at the strike.

Premium
Price of the option; ×100 per contract.

Moneyness
Price vs strike: ITM, ATM, or OTM.

Delta
Move per $1 in the stock; rough ITM odds.

Strike
The fixed price the right transacts at.

Intrinsic / Extrinsic
Real in-the-money value / time value.

Theta
Daily decay of time value.

Implied volatility
Market's expectation of movement; inflates premium.

UP NEXT IN THE PATH

Trend & Moving Averages

What MAs show, what they don't, and how to use them as context.

05